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When Should Language Models Search? A Topology-Aware Analysis of Reasoning Strategies

 (/pdf?
id=EKdCISRoGE)

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 25 May 2026 (modified: 26 Jun 2026)  ACL ARR 2026 May Submission  Everyone
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Keywords: applications, chain-of-thought

Abstract:

Search-based reasoning and chain-of-thought (CoT) prompting offer two contrasting inference-time paradigms for language models: CoT commits to a single linear trajectory, while search explores alternative partial reasoning paths before selecting an answer. In this work, we study this trade-off using A* as a controlled and optimal instantiation of heuristic-guided search. Across StrategyQA, LogicQA, HotpotQA, GSM8K, and MATH, and across three model scales, we find that search is not a universal upgrade over CoT; instead, the two strategies solve substantially different subsets of instances. To explain this complementarity, we introduce a `Reasoning Topology Framework` based on structural features computable before generation. A derived `branching coefficient` predicts when search helps better than dataset identity, indicating that the CoT--search boundary is primarily topology-dependent. We further train a topology-aware router that selects between CoT and A* without observing generated traces. Despite this restriction, it outperforms post-hoc routers based on semantic entropy and LLM-as-critic judgments. We attribute this to a `Fluency--Correctness Asymmetry`: fluent traces are often incorrect, making trace-level adjudication unreliable. Overall, our results suggest that adaptive reasoning should route by problem structure, deploying search only when the input topology warrants its higher cost. Code: `url{https://anonymous.4open.science/r/astar-reasoning-EC39%7D}` (<https://anonymous.4open.science/r/astar-reasoning-EC39%7D>).

Paper Type: Long

Research Area: Language Modeling

Research Area Keywords: chain-of-thought, prompting

Languages Studied: English

EMNLP 2026 AI Reviewing Experiment: yes

Previous URL:  /forum?id=Ze6EMFgKWZ (/forum?id=Ze6EMFgKWZ)

Explanation Of Revisions PDF:   pdf (/attachment?id=EKdCISRoGE&name=explanation_of_revisions_PDF)

Reassignment Request Area Chair:  No, I want the same area chair from our previous submission (subject to their availability).

Reassignment Request Reviewers:  No, I want the same set of reviewers from our previous submission (subject to their availability)

Preprint:  yes

Preprint Status:  We are considering releasing a non-anonymous preprint in the next two months (i.e., during the reviewing process).

Preferred Venue:  EMNLP

Visa Needs:  yes

Consent To Share Data: yes

Consent To Share Submission Details: On behalf of all authors, we agree to the terms above to share our submission details.

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: N/A

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C1 Model Size And Budget: N/A

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: Section 3.1

C3 Descriptive Statistics: Yes

C3 Elaboration: Section 3, Table 1

C4 Parameters For Packages: N/A

D Human Subjects Including Annotators: Yes

D1 Instructions Given To Participants: Yes

D1 Elaboration: Section 3.4 and Appendix B.4

D2 Recruitment And Payment: N/A

D3 Data Consent: N/A

D4 Ethics Review Board Approval: N/A

E Ai Assistants In Research Or Writing: No

E1 Information About Use Of Ai Assistants: N/A

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Submission Number: 9188

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Official Review of Submission9188 by Reviewer jRkk

Official Review by Reviewer jRkk 02 Jul 2026, 13:25 (modified: 12 Jul 2026, 21:05)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer jRkk

Paper Summary:

The authors use A* search-based reasoning to investigate how LLMs reason, whether it be through search-based or single trajectory reasoning. When investigating using several reasoning benchmarks like StrategyQA, LogicQA, HotpotQA, GSM8K, and MATH, they find that search is not always the optimal reasoning methodology, as it depends on the type of reasoning and specific task. As a result, they introduce a reasoning topology framework to determine which reasoning strategy is most effective depending on the context and question.

Summary Of Strengths:

- The authors clearly articulate the need for search-based reasoning, as they highlight the main drawbacks of CoT, including the fact that errors can propagate throughout the linear reasoning trace, negatively affecting the output.
- The experimental set-up includes several types of reasoning benchmarks, including logical, symbolic and mathematical reasoning, making the paper's evaluation comprehensive.
- Compute analysis conducted by the authors is an important part of the paper, as the authors acknowledge that A* is substantially more expensive than CoT.
- The interpretability of the proposed topology framework is the main contribution of this work, and shows how and when search can help, particularly in highly branched reasoning contexts.

Summary Of Weaknesses:

- Based on my understanding of the paper, A* uses self-reported confidence. The authors should analyze this part of the pipeline further, to identify and understand potential drawbacks.
- The human validation of the reasoning-trace quality shows a moderate agreement between moderators on a relatively small sample (100). How does this validation support the authors' work?

Comments Suggestions And Typos:

-

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3.5

Excitement: 3.5

Overall Assessment: 3.5 = Borderline Conference

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 1 = No usable datasets submitted.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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Response to weakness #1 and #2

Official Comment

by Authors ( MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

 15 Jul 2026, 22:16 (modified: 15 Jul 2026, 22:17)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer jRkk

 Revisions (/revisions?id=GOVU5Xn3zp)

Comment:

Based on my understanding of the paper, A* uses self-reported confidence. The authors should analyze this part of the pipeline further, to identify and understand potential drawbacks.

Response: Thank you for highlighting this point. The A* procedure uses self-reported confidence only in the accumulated path cost, $g(n) = \sum_t [1 + (1 - c_t)]$, rather than in the heuristic itself. The heuristic is a deterministic progress critic based on reasoning depth and question-content coverage. As stated in the manuscript, we treat c_t as a bounded search-ranking signal, not as a calibrated probability: each transition always incurs a base cost of 1, while confidence contributes only an additional penalty between 0 and 1.

This design limits, but does not eliminate, the drawbacks of self-reported confidence. An incorrect but overconfident step may be prioritized and lead to premature commitment, whereas a correct but underconfident step may be explored later. Our failure analysis captures these behaviors: among A* failures on instances correctly solved by CoT, 8.9% result from premature commitment, while 91.1% result from overthinking, where search continues beyond sufficient evidence and accumulates distractors. Thus, the dominant observed limitation is excessive exploration rather than confidence alone. Moreover, the independent depth-coverage heuristic and voting over multiple goal traces reduce dependence on any single confidence estimate. Accordingly, our claims concern this controlled A* instantiation and do not assume that self-reported confidence is calibrated or universally reliable.

W2. The human validation of the reasoning-trace quality shows a moderate agreement between moderators on a relatively small sample (100). How does this validation support the authors' work?

Response: Thank you for raising this question. The human evaluation contains 100 CoT-A* disagreement instances from each of StrategyQA and LogicQA, for 200 annotated instances in total, selected from disagreement pools of 587 and 332 instances, respectively. The purpose of this study is not to establish aggregate performance, which is supported by the full-benchmark accuracy and instance-level decomposition, but to diagnose why the two strategies succeed on different examples.

The results show the same pattern across both datasets and in both directions. When CoT is correct and A* is wrong, CoT has substantially higher reasoning coverage than A*: 3.00 versus 1.20 on StrategyQA and 3.00 versus 1.86 on LogicQA. Conversely, when A* is correct, A* has higher coverage: 2.71 versus 1.63 on StrategyQA and 2.77 versus 1.88 on LogicQA. Relevance is much less strongly associated with correctness, showing that a trace can remain on-topic while still missing the decisive inferential steps.

We interpret the moderate agreement, $\kappa = 0.59\text{--}0.68$, as reflecting the inherent subjectivity of ordinal trace-quality assessment, and therefore do not use the human scores as primary statistical evidence. Rather, their consistent directional pattern supports the narrower mechanistic conclusion that CoT-A* complementarity is driven mainly by whether the strategy covers the required reasoning path, rather than by superficial fluency or stylistic quality.

Add: **Author-Editor Confidential Comment**

Paper Summary:

This paper analyzes when language models should use Chain-of-Thought (CoT) reasoning and when they should use search-based reasoning during inference. The authors use A* search as a controlled search method and compare it with CoT, showing that neither strategy uniformly outperforms the other; instead, they are complementary and solve different types of problems. To explain this complementarity, the paper introduces reasoning topology features and a branching coefficient that characterize the structural properties of each problem. Based on these features, the authors train a topology-aware router that selects between CoT and A* before generation. The experimental results suggest that search is more beneficial for multi-hop or branch-heavy problems, whereas CoT can be more suitable for simpler, more linear problems.

Summary Of Strengths:

1. The research question is clear and practical. Rather than simply asking whether search is better than CoT, the paper addresses the more useful question of when search is actually needed. This is an important issue for inference-time reasoning, where both accuracy and computational cost must be considered.
2. The paper analyzes the complementarity between CoT and search at the instance level. Across multiple benchmarks and model scales, the authors show that CoT and A* solve different subsets of problems, and the oracle gap makes a strong case for routing between reasoning strategies. This goes beyond average accuracy comparisons and helps identify which types of problems benefit from each strategy.
3. The paper proposes an interpretable and interesting direction through topology-aware routing. Instead of evaluating reasoning traces after generation, the authors select a reasoning strategy based only on the problem structure before generation. The result that a router using branching coefficients and topology features outperforms post-hoc trace-based routers is compelling and provides a useful perspective for adaptive reasoning.

Summary Of Weaknesses:

1. **There appears to be an unexplained discrepancy between the aggregate accuracy in Table 1 and the instance-level decomposition in Table 2.** In Table 2, “CoT only” is defined as cases where CoT is correct and A* is incorrect, while “A* only” is defined as the opposite. Therefore, if Tables 1 and 2 are based on the same evaluation set, it would be natural to calculate CoT accuracy as “Both correct + CoT only” and A* accuracy as “Both correct + A* only” for each dataset. However, for StrategyQA, if calculated based on Table 2, the CoT accuracy should be approximately 73.8% and the A* accuracy approximately 70.3%, whereas Table 1 reports CoT as 64.52% and A* as 74.80%. In LogicQA and HotpotQA as well, the A* accuracy differs between the values calculated from Table 2 and those reported in Table 1. If this discrepancy is due to different evaluation subsets, filtering, or aggregation procedures, it should be clearly explained; otherwise, it could undermine the credibility of the paper’s core quantitative results.
2. **The term “topology” and the definition of the features are somewhat ambiguous.** The topology features proposed in the paper are closer to hand-crafted structural or input features—such as question length, context length, logical connectives, negation markers, estimated hop count, distractor count, and number of answer options—than to mathematical topology or TDA-style features. A clearer definition and justification are needed to explain why these features should be considered to capture reasoning topology. Furthermore, the claim would be more persuasive if the paper further verified that these features reflect actual reasoning structures rather than dataset artifacts or surface statistics.
3. **The rationale for choosing A* search and its relationship to other search methods are not sufficiently clear.** Although the paper explains that A* is used as a controlled diagnostic, the justification for why A* is more appropriate for this research question than other search-based reasoning methods—such as beam search, Tree-of-Thought, and MCTS-style search—is somewhat limited. In particular, since the paper’s conclusion extends to the general question of when search is helpful, there should be a clearer discussion of whether the topology-dependent behavior observed with A* would also hold for other search algorithms, or whether it depends on the specific heuristic, frontier management, and cost function used by A*. The paper’s generalizability would be strengthened if ablation studies with other search methods, or at least a more detailed comparative analysis, were included.

4. **The scope and role of human evaluation are limited.** While the evaluation of some disagreement instances by two annotators is meaningful as a supplementary analysis, it is somewhat limited in supporting the broader claims about the relationship between trace fluency and correctness and the superiority of topology-aware routing. Human validation involving more annotators, a larger sample size, and more diverse datasets would improve the reliability of the analysis of the Fluency-Correctness Asymmetry and routing behavior.

Comments Suggestions And Typos:

Minor issue: Since the branching coefficient is one of the paper's core concepts, it would be more appropriate for readability and clarity to present its definition in a separate equation block rather than only in inline text.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 2.5 = Borderline Findings

Ethical Concerns:

There are no concerns with this submission

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

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Response to #W1

Official Comment

by Authors (MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

12 Jul 2026, 14:05

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer qkS1, Authors

Comment:

W1. There appears to be an unexplained discrepancy between the aggregate accuracy in Table 1 and the instance-level decomposition in Table 2. In Table 2, "CoT only" is defined as cases where CoT is correct and A* is incorrect, while "A* only" is defined as the opposite. Therefore, if Tables 1 and 2 are based on the same evaluation set, it would be natural to calculate CoT accuracy as "Both correct + CoT only" and A* accuracy as "Both correct + A* only" for each dataset. However, for StrategyQA, if calculated based on Table 2, the CoT accuracy should be approximately 73.8% and the A* accuracy approximately 70.3%, whereas Table 1 reports CoT as 64.52% and A* as 74.80%. In LogicQA and HotpotQA as well, the A* accuracy differs between the values calculated from Table 2 and those reported in Table 1. If this discrepancy is due to different evaluation

subsets, filtering, or aggregation procedures, it should be clearly explained; otherwise, it could undermine the credibility of the paper's core quantitative results.

Response to W1. Thank you for identifying this inconsistency. The reviewer is correct that, when Tables 1 and 2 are computed over the same paired evaluation instances, the marginal accuracies must satisfy $Acc_{CoT} = (N_{both} + N_{CoT-only})/N$ and $Acc_{A^*} = (N_{both} + N_{A^*-only})/N$. Similarly, the instance-wise oracle accuracy must satisfy $Acc_{oracle} = (N_{both} + N_{CoT-only} + N_{A^*-only})/N$.

The discrepancy resulted from a manuscript-versioning error. Table 2 was inadvertently retained from an earlier A^* configuration, whereas Table 1 reports results obtained using the final depth--coverage heuristic. The discrepancy was therefore not caused by different evaluation subsets, filtering rules, or aggregation procedures.

We have now recomputed the instance-level decomposition using the same paired CoT and final A^* prediction files used to produce Table 1. The corrected Table 2 is shown below.

Dataset	Both correct	CoT only	A^* only	Both wrong
StrategyQA	54.73%	9.79%	20.07%	15.41%
LogicQA	31.34%	14.44%	13.51%	40.71%
HotpotQA	62.70%	14.10%	17.40%	5.80%

The corrected decomposition exactly reproduces the aggregate results at the precision reported in the manuscript:

- **StrategyQA:** CoT accuracy is $54.73 + 9.79 = 64.52$, A^* accuracy is $54.73 + 20.07 = 74.80$, and oracle accuracy is $54.73 + 9.79 + 20.07 = 84.59$.
- **LogicQA:** CoT accuracy is $31.34 + 14.44 = 45.78$, A^* accuracy is $31.34 + 13.51 = 44.85$, and oracle accuracy is $31.34 + 14.44 + 13.51 = 59.29$.
- **HotpotQA:** CoT accuracy is $62.70 + 14.10 = 76.80$, A^* accuracy is $62.70 + 17.40 = 80.10$, and oracle accuracy is $62.70 + 14.10 + 17.40 = 94.20$.

For every dataset, the four categories also sum to 100. Therefore, the aggregate results in Table 1 remain unchanged, while the corrected decomposition continues to demonstrate substantial instance-level complementarity between CoT and A^* .

We will replace Table 2 with the corrected table above and revise the corresponding discussion.

We apologize for this version mismatch and thank the reviewer for bringing it to our attention.

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Response to #W2

Official Comment

by Authors (👤 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

📅 12 Jul 2026, 14:07

👤 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer qkS1, Authors

Comment:

The term “topology” and the definition of the features are somewhat ambiguous. The topology features proposed in the paper are closer to hand-crafted structural or input features—such as question length, context length, logical connectives, negation markers, estimated hop count, distractor count, and number of answer options—than to

mathematical topology or TDA-style features. A clearer definition and justification are needed to explain why these features should be considered to capture reasoning topology. Furthermore, the claim would be more persuasive if the paper further verified that these features reflect actual reasoning structures rather than dataset artifacts or surface statistics.

Response to W2: Thank you for this observation. We agree that *reasoning topology* does not refer to mathematical topology or topological data analysis. We use the term operationally to describe the structural form of reasoning induced by an input: whether the problem can be solved through a largely linear and globally coherent trajectory, or instead requires preserving, comparing, or eliminating multiple partial reasoning paths before committing to an answer.

This interpretation follows the qualitative regimes identified in Section 3.2 and Appendix B. In particular, A^* is most useful for **non-linear evidence aggregation**, **hypothesis elimination**, and **multi-hop composition**, where several candidate facts, constraints, or reasoning paths must be considered before commitment. In contrast, CoT is preferable for **direct factual reasoning** and **globally coherent argumentation**, where a single continuous reasoning trajectory is sufficient.

We agree that the 13 proposed features are not topological invariants in the mathematical sense. Rather, they are **pre-generation reasoning-topology proxies** intended to capture properties associated with reasoning depth, branching opportunity, competing constraints, and the amount of evidence that must be integrated. For example, the branching coefficient is defined as:

$$bc = (c_sent \times hop) / q_len$$

where c_sent is the number of context sentences, hop is the estimated reasoning-hop count, and q_len is the question length. This quantity combines contextual extent and estimated reasoning depth while normalizing for question length. It is therefore intended as an operational proxy for the amount of multi-hop branching structure afforded by an input, rather than as a formal topological quantity.

The current results provide several complementary pieces of evidence that the observed relationship is not explained purely by dataset identity. First, each of the four reasoning-topology clusters contains instances from multiple benchmarks, while the A^* advantage increases from -0.4 percentage points in the **linear, short** cluster to +11.4 percentage points in the **deep-branch** cluster. The relationship between the branching coefficient and A^* advantage across clusters has $R^2 = 0.94$. Second, the complementary topic analysis finds that A^* -dominant categories are primarily associated with multi-hop entity chains, whereas CoT-dominant categories are primarily associated with single-step commonsense reasoning. Third, the human analysis indicates that disagreement outcomes are better explained by whether a method covers the required reasoning path than by the apparent fluency or surface quality of its trace.

Nevertheless, we agree that these analyses do not completely eliminate the possibility that the handcrafted proxies retain some surface-level or dataset-specific signal. We will therefore revise the manuscript to:

1. explicitly define *reasoning topology* as the branching, depth, and path-comparison structure induced by a reasoning problem;
2. refer to the proposed features as **reasoning-topology proxies**, rather than formal topology features;
3. clarify that the cluster, topic, and human analyses provide supporting evidence against a purely dataset-specific explanation, rather than definitive proof that all surface-statistic confounds have been eliminated; and
4. acknowledge more explicitly that some proxies, including input length and connective counts, may partially capture surface properties in addition to reasoning structure.



➔ *Replying to Response to #W2*

Response to #W3

Official Comment

by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

📅 12 Jul 2026, 14:27

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer qkS1, Authors

Comment:

The rationale for choosing A* search and its relationship to other search methods are not sufficiently clear. Although the paper explains that A* is used as a controlled diagnostic, the justification for why A* is more appropriate for this research question than other search-based reasoning methods—such as beam search, Tree-of-Thought, and MCTS-style search—is somewhat limited. In particular, since the paper’s conclusion extends to the general question of when search is helpful, there should be a clearer discussion of whether the topology-dependent behavior observed with A* would also hold for other search algorithms, or whether it depends on the specific heuristic, frontier management, and cost function used by A*. The paper’s generalizability would be strengthened if ablation studies with other search methods, or at least a more detailed comparative analysis, were included.

Response to W3: We use A* not because it is always better than beam search, Tree-of-Thought, or MCTS, but because it offers a clean way to study heuristic-guided search. Its design separates path cost, heuristic estimate, frontier management, and branching. This makes it easier to examine the main contrast with CoT: exploring and prioritizing multiple partial reasoning paths instead of committing to a single linear trace.

Other search methods introduce additional design choices. Beam search uses fixed-width pruning, Tree-of-Thought relies on an evaluator, and MCTS depends on rollout, value estimation, exploration, and backup rules. These are important methods, but their extra components make it harder to isolate why search helps.

We therefore do not claim that all search algorithms will show the same gains or the same routing boundary. Our claim is more limited: problem structure determines whether there is room for search to help, while the search algorithm determines how well that opportunity is used. Problems involving multi-hop composition, hypothesis elimination, or competing evidence paths are more likely to benefit from methods that delay commitment. However, A*, beam search, Tree-of-Thought, and MCTS may differ in pruning errors, search efficiency, and overthinking.

Our analysis thus has two levels. At the structural level, we study when maintaining multiple partial reasoning paths is better than following one linear path. This idea is shared by several search methods. At the algorithmic level, the methods differ in how they prune, score, explore, and allocate budget. Therefore, we expect topology to matter across search methods, but not necessarily to produce the same gains or the same routing threshold. The boundary reported in this paper is specific to our budgeted A* implementation. Testing whether the same pattern holds for beam search, Tree-of-Thought, and MCTS is an important direction for future work.



➔ *Replying to Response to #W3*

Response to W4

Official Comment

by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

📅 12 Jul 2026, 14:29

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer qkS1, Authors

Comment:

The scope and role of human evaluation are limited. While the evaluation of some disagreement instances by two annotators is meaningful as a supplementary analysis, it is somewhat limited in supporting the broader claims about the relationship between trace fluency and correctness and the superiority of topology-aware routing. Human validation involving more annotators, a larger sample size, and more diverse datasets would improve the reliability of the analysis of the Fluency-Correctness Asymmetry and routing behavior.

Response to W4. Thank you for raising this concern. We agree that the current human evaluation should be interpreted as a supplementary diagnostic analysis rather than as comprehensive human validation across all tasks and routing settings.

We would first like to clarify that the empirical superiority of the topology-aware router is not inferred from the human evaluation. It is measured directly on the complete evaluation sets of five benchmarks. As reported in Table 4, the topology-aware router outperforms CoT, A*, semantic-entropy routing, LLM-as-Critic, LLM-as-Judge, and the random-forest router on StrategyQA, LogicQA, HotpotQA, GSM8K, and MATH. The improvements of the topology router over CoT on the three QA benchmarks are also statistically significant under McNemar's test with $p < 0.005$. Thus, the routing result is based on benchmark-level predictive accuracy, whereas the human evaluation is used only to provide an interpretation of why retrospective trace comparison can be unreliable.

The human analysis considers 100 sampled CoT--A* disagreement instances from StrategyQA and 100 from LogicQA. Two postgraduate annotators with experience in language-model reasoning evaluated both traces in each instance. Inter-annotator agreement ranges from Cohen's kappa of 0.59 to 0.68 across the reported criteria. The annotations consistently show that *the correct strategy obtains substantially higher coverage of the required reasoning path than the incorrect strategy*. For example, on StrategyQA, CoT receives an average coverage score of 3.00 versus 1.20 for A* when CoT is correct, while A* receives 2.71 versus 1.63 for CoT when A* is correct. The same directional pattern is observed on LogicQA.

We nevertheless agree that two annotators, two datasets, and 200 disagreement cases are insufficient to establish a universal relationship between trace fluency and correctness. We will therefore revise the manuscript to make the scope of this evidence explicit. In particular, we will:

1. Describe the human evaluation as a supplementary analysis of sampled disagreement cases rather than as validation of the routing result itself; and we will also use all datasets and three annotators and atleast 500 examples per dataset.
2. State that the observed Fluency--Correctness Asymmetry is an empirical pattern in the analyzed disagreement cases, rather than a universal property of all reasoning traces;

Accordingly, our revised conclusion will be that the benchmark-level results establish the advantage of topology-aware routing in the evaluated setting, while the human analysis provides limited but consistent supporting evidence that apparent trace quality does not always reveal whether the decisive reasoning path has been covered.

Add: **Author-Editor Confidential Comment**



➔ *Replying to Response to W4*

Official Comment by Reviewer qkS1

Official Comment by Reviewer qkS1 📅 13 Jul 2026, 09:41

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer qkS1, Authors

Comment:

Thank you for the detailed response. The corrected instance-level decomposition and the explanation of the manuscript-versioning issue substantially address my primary concern in W1 regarding the numerical inconsistency between Tables 1 and 2. The clarifications and planned revisions for W2–W4 also sufficiently strengthen the technical presentation, terminology, and scope of the claims. I am therefore raising my score from 2.0 to 2.5. The magnitude of the inconsistency in the initial submission still limits a larger upward revision, although the rebuttal has meaningfully improved my overall assessment of the work.

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Official Review of Submission9188 by Reviewer s1To

Official Review by Reviewer s1To 📅 27 Jun 2026, 12:19 (modified: 12 Jul 2026, 21:05)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer s1To

📄 Revisions (/revisions?id=ovYhi70FAD)

Paper Summary:

The paper studies when language models should use search-based reasoning instead of standard chain-of-thought prompting. It compares CoT with a budgeted A* search procedure over natural-language reasoning traces. Across StrategyQA, LogicQA, HotpotQA, GSM8K, and MATH, the paper finds that neither CoT nor A* uniformly dominates. The paper further proposes a topology-aware router based on 13 pre-generation structural features, arguing that whether search helps is better explained by problem topology than by dataset identity.

Summary Of Strengths:

- Deciding when to spend additional inference-time compute is important, especially as search-based reasoning methods become more common.
- The instance-level complementarity analysis is useful. The paper shows that CoT and A* solve different subsets of examples, which supports the basic motivation that search and linear reasoning are not simply ordered by overall accuracy.
- The paper provides useful analysis of when A* helps and when it hurts. The discussion around deferred commitment, overthinking, and fluency-correctness asymmetry helps explain why search is not always better than CoT.

Summary Of Weaknesses:

- The topology-aware routing result needs more careful explanation. The router is trained on 13 manually designed features, but the paper does not make the training/evaluation protocol clear enough in the main text. Since the main claim is that routing can be done from input structure before generation, the paper should clearly

describe the train/test split, whether splits are within-dataset or cross-dataset, and whether there is any benchmark-format leakage.

- The routing setup is somewhat narrow. The router mainly chooses between CoT and A*, but Table 1 shows that self-consistency is often competitive, and sometimes stronger, especially on StrategyQA, LogicQA, GSM8K, and MATH. If the paper asks when language models should search, the routing decision should ideally include strong non-search inference strategies, not only CoT versus A*.
- The compute cost remains important for the main claim. The paper reports that default A* uses around 12,452 tokens per question, about 24× CoT. Since search is much more expensive, the paper should make the routing objective explicitly cost-aware rather than only accuracy-oriented.
- Some terminology is confusing. The paper uses “critic” in multiple ways: a rule-based progress critic, an LLM-as-critic router, and an LLM-as-judge baseline. These should be separated more clearly because they play different roles in the method.
- The actual implementation of A* is not simple. It has many components, but the paper only ablates budget and branching factor. It does not isolate key choices such as confidence-based cost, depth/coverage heuristics, successor generation, or goal voting, so it is unclear which parts actually drive the gains.

Comments Suggestions And Typos:

- It would be helpful to test whether the topology router is learning general reasoning structure or benchmark-specific formatting artifacts.
- Since self-consistency is competitive in Table 1, the paper should explain why routing is only between CoT and A*.
- Including more diverse/recent models would definitely be preferred.
- For cost, it is useful to compare cost across all the methods, not just between A* and CoT to better situate how A*.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 2.5

Excitement: 2.5

Overall Assessment: 2.5 = Borderline Findings

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

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Response to #W1, #W2

Official Comment

by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

Comment:

W1. The topology-aware routing result needs more careful explanation. The router is trained on 13 manually designed features, but the paper does not make the training/evaluation protocol clear enough in the main text. Since the main claim is that routing can be done from input structure before generation, the paper should clearly describe the train/test split, whether splits are within-dataset or cross-dataset, and whether there is any benchmark-format leakage.

Response to W1. Thank you for pointing out that the router evaluation protocol was not described clearly enough in the main text. We agree that this information is necessary to interpret the topology-routing result correctly.

The results in Table 4 use stratified 5-fold cross-validation with seed 42, performed separately within each benchmark. For each fold, the router is trained on four folds and evaluated only on the held-out fold. Thus, no evaluation instance is used to train the router that predicts it. The reported accuracy for each benchmark is obtained by combining the predictions from its five held-out folds.

The router receives only the 13 features computed from the input before generation. It does not observe the CoT trace, the A* trace, model confidence, predicted answers, correctness labels, or gold answer at inference time. Dataset names are also not provided as features. The training target is derived from the paired CoT and A* outcomes and indicates which strategy should be selected for that instance. We will explicitly state how ties are handled when both strategies are correct or both are incorrect.

We also clarify that this protocol evaluates **within-dataset generalization**, not transfer to an unseen benchmark. Therefore, the current result supports the claim that input-side structural features can predict the preferred strategy for unseen instances from the same benchmark distribution. It does not by itself establish that a router trained on some benchmarks will generalize to a new benchmark.

The reviewer is also correct that the absence of an explicit dataset identifier does not completely rule out benchmark-format information. Features such as context length, number of context sentences, and number of answer options may indirectly encode dataset format. We will therefore revise the manuscript to distinguish reasoning-structure features from format-sensitive features and avoid claiming that the current experiment completely eliminates dataset-specific signal.

In the revision, we will add the following details to the main text:

- the stratified 5-fold within-dataset evaluation protocol and random seed;
- the construction of the routing target, including the treatment of ties;
- confirmation that feature extraction and preprocessing are performed without using generated traces or evaluation labels at inference time.

W2. The routing setup is somewhat narrow. The router mainly chooses between CoT and A*, but Table 1 shows that self-consistency is often competitive, and sometimes stronger, especially on StrategyQA, LogicQA, GSM8K, and MATH. If the paper asks when language models should search, the routing decision should ideally include strong non-search inference strategies, not only CoT versus A*.

Response to W2. Thank you for raising this point. We agree that the current routing experiment addresses a narrower question than general inference-strategy selection. Our binary router was designed to isolate the central comparison studied in the paper: whether an instance should be solved using a single linear CoT trajectory or using heuristic-guided search over multiple partial reasoning states. Self-consistency differs from A* in an important way: it samples several independent complete CoT trajectories and aggregates their answers, but it does not maintain a search frontier, expand partial states, backtrack, or prioritize branches using a heuristic. Therefore, the current CoT--A* router specifically tests whether the input structure predicts the value of explicit branching and deferred commitment. Nevertheless, the reviewer is correct that self-consistency is a strong inference-time alternative. In Table 1, it is competitive

with A* on StrategyQA and HotpotQA and achieves the highest LLaMA-3.1-8B accuracy among the fixed methods on LogicQA, GSM8K, and MATH. Consequently, **the current binary router should not be interpreted as identifying the globally best inference strategy among CoT, self-consistency, and A*. It establishes only that pre-generation input features can help decide between linear CoT and our budgeted A* search.**

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Response to #W3, #W4, #W5

Official Comment
by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))
📅 12 Jul 2026, 17:38 (modified: 12 Jul 2026, 17:39)
👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer s1To, Authors
📄 Revisions (/revisions?id=gPHt0mteyn)

Comment:
W3. The compute cost remains important for the main claim. The paper reports that default A* uses around 12,452 tokens per question, about 24× CoT. Since search is much more expensive, the paper should make the routing objective explicitly cost-aware rather than only accuracy-oriented.

Response: Thank you for highlighting this point. We agree that a deployment-oriented router should consider both accuracy and inference cost. However, the purpose of the current routing experiment is to test a more specific question: whether the relative accuracy advantage of CoT and A* can be predicted from the input structure before either reasoning procedure is executed. Accordingly, the current router is accuracy-oriented by design. This isolates the main scientific claim of the paper: problem topology predicts when explicit branching is useful. A cost-weighted objective would introduce an additional deployment-dependent trade-off parameter and could obscure whether the CoT--A* boundary is structurally predictable in the first place.

W4. Some terminology is confusing. The paper uses “critic” in multiple ways: a rule-based progress critic, an LLM-as-critic router, and an LLM-as-judge baseline. These should be separated more clearly because they play different roles in the method.

Response: Thank you for pointing out this terminology overlap. We agree that the word *critic* is currently used for components with different roles.

The three components are functionally distinct:

Component	Role	Input	Used during
Rule-based progress critic	Computes the A* heuristic score	A partial reasoning state	A* search
LLM-as-Critic	Selects between completed CoT and A* traces	Both full reasoning traces	Post-hoc routing
LLM-as-Judge	Alternative LLM-based trace-selection baseline	Both full reasoning traces	Post-hoc routing

The rule-based progress critic is not an LLM and does not evaluate factual correctness. It is a deterministic scoring function based on reasoning depth and question-content coverage, and it is used only to prioritize partial states within A*. In contrast, LLM-as-Critic and LLM-as-Judge are

output-based routing baselines that inspect completed CoT and A* traces and select between them. These two baselines differ only in their prompting formulation.

We will revise the terminology to make these roles clearer. Specifically, we will rename the **rule-based progress critic** as the **rule-based progress scorer** throughout the paper. We will also describe LLM-as-Critic more explicitly as a **pairwise LLM trace selector**, while retaining LLM-as-Judge as a separate prompting baseline. In addition, we will add a short clarification in Section 2.5 distinguishing search-time scoring from post-hoc trace selection.

This revision is terminological and does not change the method or experimental results.

W5. The actual implementation of A* is not simple. It has many components, but the paper only ablates budget and branching factor. It does not isolate key choices such as confidence-based cost, depth/coverage heuristics, successor generation, or goal voting, so it is unclear which parts actually drive the gains.

Response: Thank you for this point. We agree that the current ablations do not fully attribute the gains to individual components of the A* implementation. Our budget and branching-factor studies isolate the main search-capacity variables, but not the effects of the confidence-based path cost, depth/coverage heuristic, successor generation, or goal voting.

Our claim is therefore not that each component is independently necessary, but that this complete budgeted A* instantiation provides a controlled test of heuristic-guided branching versus linear CoT. We will clarify this scope and add targeted ablations where feasible, including removing the confidence penalty, separating the depth and coverage heuristic terms, and replacing multi-goal voting with single-best-goal selection. We will also identify full component attribution as a limitation of the current study.

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➡ *Replying to Response to #W3, #W4, #W5*

Response to Comments Suggestions And Typos

Official Comment

by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

📅 12 Jul 2026, 18:19

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer s1To, Authors

Comment:

Response to Comment 1. Thank you for raising this concern. We agree that excluding dataset IDs does not fully prevent benchmark leakage, since structural features may still reveal the dataset.

We have performed additional analyses across pooled benchmarks, task domains, and model scales. First, the topology clusters are constructed over pooled QA and mathematics instances, with each cluster containing examples from multiple benchmarks. Despite this mixing, the relative advantage of A* increases with the branching coefficient:

Cluster	Structural profile	Branching coefficient	CoT	A*	A* advantage
C1	Linear, short	1.28	82.1	81.7	-0.4
C3	Complex, noisy	4.02	49.1	52.3	+3.2
C2	Mid-branch	4.24	66.2	73.5	+7.3

Cluster	Structural profile	Branching coefficient	CoT	A*	A* advantage
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C4	Deep-branch	6.49	58.4	69.8	+11.4
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The relationship between branching coefficient and A* advantage has an R^2 of 0.94. Because the clusters contain instances from multiple benchmarks, this pattern cannot be explained simply by assigning one strategy to each dataset.

Second, we have evaluated the same 13-feature routing formulation across eight benchmarks spanning QA, mathematics, and code generation, and across three model scales:

Domain	Benchmark	Qwen-0.5B	LLaMA-3.1-8B	Qwen-2.5-14B
QA	StrategyQA	64.74	77.35	76.46
QA	LogicQA	24.98	49.01	58.68
QA	HotpotQA	30.00	82.41	83.17
Mathematics	GSM8K	26.84	91.10	80.82
Mathematics	MATH	33.21	81.52	48.40
Code	HumanEval	34.97	72.50	79.75
Code	MBPP	25.99	40.63	48.42
Code	CodeContests	75.20	95.00	--

The values are topology-router accuracies. The router remains effective across substantially different formats, including boolean QA, multiple-choice logical reasoning, mathematical problems, code generation, and programming. We also performed an independent cross-dataset topic analysis over 3,660 questions: A* is stronger in 6 of 8 topic categories covering 66.2% of the instances, and these categories primarily involve multi-hop factual or entity chains; the two CoT-dominant categories primarily involve single-step commonsense reasoning.

These results provide evidence against a purely benchmark-specific explanation, but we agree that they do not completely eliminate indirect format leakage. We will therefore distinguish **reasoning-structure features** from **format-sensitive features** more clearly and avoid presenting the current results as definitive evidence of unseen-benchmark transfer. We will also add an ablation removing raw length and answer-format features and report leave-one-benchmark-out routing, which directly tests whether the learned decision rule transfers to an unseen benchmark format.

Response to Comment 2. Thank you for this point. We agree that self-consistency is a strong inference-time alternative. We have compared CoT, A*, self-consistency, and Tree-of-Thought across all eight benchmarks. The LLaMA-3.1-8B results are:

Benchmark	CoT	A*	Self-Consistency	Tree-of-Thought
StrategyQA	64.52	74.80	71.22	72.24
LogicQA	45.78	44.85	46.10	46.21
HotpotQA	76.80	80.10	78.21	78.45
GSM8K	85.04	86.12	86.90	60.00
MATH	78.00	79.10	79.50	40.10
HumanEval	67.48	64.42	65.03	71.78
MBPP	39.80	36.80	38.00	40.40

Benchmark	CoT	A*	Self-Consistency	Tree-of-Thought
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CodeContests	77.40	87.00	99.00	91.20
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These results confirm the reviewer's observation: self-consistency is strongest on GSM8K, MATH, and CodeContests, and remains competitive on several other benchmarks.

The present binary router is deliberately designed to isolate the specific comparison studied in the paper: a single linear CoT trajectory versus A* search over partial reasoning states. Self-consistency samples multiple independent complete CoT traces and aggregates their answers, but it does not maintain a frontier, expand partial states, backtrack, or prioritize branches with a heuristic. It therefore captures sampling diversity, whereas A* captures guided exploration and deferred commitment.

Accordingly, the current routing result answers the narrower question: **when should explicit heuristic-guided search replace single-trajectory CoT?** It should not be interpreted as identifying the globally best inference strategy among CoT, self-consistency, A*, and Tree-of-Thought. We will revise the paper to state this scope explicitly and refer to the current model as a *CoT-A search router**, rather than a general inference-strategy router.

The results above also motivate a broader cost-aware router over {CoT, self-consistency, A*}, which would choose among single-path reasoning, sampling-based aggregation, and explicit search. We will identify this as an important extension rather than implying that the current binary router covers all inference-time strategies.

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➔ *Replying to Response to Comments Suggestions And Typos*

Response to comment 3,4

Official Comment
by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))
📅 12 Jul 2026, 18:23
👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer s1To, Authors

Comment:
Response to Comment 3. Thank you for these suggestions. We agree that broader model coverage and a more complete cost comparison are important for establishing the scope of the conclusions.

We have performed experiments with two model families and three substantially different scales: Qwen-0.5B, LLaMA-3.1-8B, and Qwen-2.5-14B. We evaluate each model across eight benchmarks spanning QA, mathematics, and code generation. Representative CoT and A* results are shown below:

Model	StrategyQA: CoT / A*	HotpotQA: CoT / A*	GSM8K: CoT / A*	CodeContests: CoT / A*
Qwen-0.5B	53.60 / 53.40	8.40 / 30.00	26.84 / 14.48	49.80 / 68.20
LLaMA-3.1-8B	64.52 / 74.80	76.80 / 80.10	85.04 / 86.12	77.40 / 87.00
Qwen-2.5-14B	76.02 / 75.56	60.19 / 83.33	61.56 / 80.59	99.60 / 94.40

The results show that the effect of search is not tied to one model family or scale. For example, A* improves HotpotQA by 21.60 points at 0.5B and 23.14 points at 14B, improves GSM8K by 19.03 points at 14B, and improves CodeContests by 18.40 points at 0.5B and 9.60 points at 8B. At the same time, A* can underperform on structurally simpler tasks, reinforcing the need for routing. We agree that evaluation on additional recent model families would further strengthen generality, and we will state the current model coverage clearly rather than implying universality.

Response to Comment 4. Thank you for these suggestions. We have also performed a broader compute comparison on the same stratified 300-question HotpotQA sample. All token counts below are measured rather than estimated:

Method	Accuracy	Tokens per question	Relative to CoT
CoT	76.80	approximately 512	1.0x
Self-Consistency, 5 traces	78.21	approximately 2,560	5.0x
A*, K=1, B=30	77.70	3,372	6.6x
A*, K=3, B=15	78.00	12,452	24.3x
A*, K=3, B=20	78.70	12,391	24.2x

This comparison places A* more clearly relative to a strong multi-sample baseline. Reduced-budget A* is moderately more expensive than five-trace self-consistency, while default A* is substantially more expensive for a relatively small additional accuracy gain on this sample. This supports our central motivation for pre-generation routing: expensive search should be reserved for instances whose structure indicates that branching is likely to help.

We agree that a complete comparison should also include measured end-to-end costs for Tree-of-Thought and the routing baselines. In particular, semantic-entropy, LLM-as-Critic, and LLM-as-Judge routing require generating multiple candidate traces before routing, and the LLM-based selectors require an additional model call. By contrast, the topology router uses only input-side features and requires no LLM call before executing the selected strategy. We will report the measured end-to-end token costs of these methods so that their accuracy can be compared under the same computational accounting.

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➤ *Replying to Response to comment 3,4*

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by Authors (👁 MD Tanzeel Adam Khan (/profile?id=~MD_Tanzeel_Adam_Khan1), Dibyanayan Bandyopadhyay (/profile?id=~Dibyanayan_Bandyopadhyay1), Asif Ekbal (/profile?id=~Asif_Ekbal1))

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer s1To

Comment:
We thank reviewer S1To for the detailed review of our manuscript. We have carefully addressed each of the points raised above. We kindly request you to consider our responses and, if appropriate, update your score.

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FAQ (https://docs.openreview.net/getting-started/frequently-asked-questions)

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